Industrial SO2 and NO2 watch: Maharashtra

Udyog-drishti AI · Sentinel-5P · 2019-01 to 2026-09

Item	Value
Sites	13 coal power sites (22,020 MW) and 4 industrial areas
Check first / watch	5 / 6
Reference plants with scrubbers	Mauda (NTPC), Dahanu (Adani), Trombay–Chembur
Months analysed	93

Method

Sites: Coal plants of 500 MW or more from the Global Power Plant Database, plus three CPCB critically polluted clusters and Chandrapur's cement belt. Plants closer than ~10 km are one site: the satellite cannot separate them.

Enhancement: Monthly mean column within 10 km of the site minus the median in a 40–80 km ring (regional background). TROPOMI offline L3, cloud fraction under 0.3. Units: µmol/m².

Years: Years are compared on January–May only: clear skies and no monsoon washout.

Status: Check first: SO2 above background of 150+ µmol/m² over the last 12 months, or rising 25%+ against 2021-2024 (January–May). Watch: 40+ µmol/m². Low: below that, within the noise of a monthly mean.

Scrubber: SO2 ÷ NO2 above background, before vs after a date, minus the same change at exempt plants without scrubbers. A scrubber cuts SO2 but not NO2.

Limits: Columns, not stack concentrations: no verdict on the mg/Nm³ norm. One overpass a day (~13:30), so night releases are missed. Sites a few km apart are merged. Retrieval changes and coal sulphur content also move the numbers, so a change is a reason to inspect, not proof.

Results

Site	Status	Why	Type	District	Capacity (MW)	SO2 above background, last 12 months (µmol/m²)	NO2 above background, last 12 months (µmol/m²)	SO2 ÷ NO2	SO2 change, latest Jan–May vs 2021-2024	Scrubber (FGD)
Chandrapur STPS + Dhariwal	check first	high SO2	Coal power	Chandrapur	3940	238.2	95.0	2.51	-0.176	planned
Koradi + Khaparkheda	check first	high SO2	Coal power	Nagpur	3740	189.9	85.9	2.21	-0.269	planned
Warora (GMR + Sai Wardha)	check first	high SO2	Coal power	Chandrapur	1140	169.0	34.8	4.86	0.004	no
Tiroda (Adani)	check first	high SO2	Coal power	Gondia	3300	161.1	32.8	4.91	-0.292	no
Ghugus–Rajura cement belt	check first	SO2 rising	Cement cluster	Chandrapur		98.3	42.6	2.31	0.285	
Mauda (NTPC)	watch		Coal power	Nagpur	2320	132.5	51.2	2.59	-0.531	yes
Bhusawal (Deepnagar)	watch		Coal power	Jalgaon	1420	77.1	28.4	2.71	-0.038	

Site	Status	Why	Type	District	Capacity (MW)	SO2 above background, last 12 months (µmol/m²)	NO2 above background, last 12 months (µmol/m²)	SO2 ÷ NO2	SO2 change, latest Jan–May vs 2021-2024	Scrubber (FGD)
Amravati (RattanIndia)	watch		Coal power	Amravati	1350	74.8	28.0	2.67	-0.085	no
Butibori	watch		Coal power	Nagpur	600	68.8	29.2	2.36		no
Parli	watch		Coal power	Beed	1380	52.7	17.2	3.06	-0.042	
Trombay–Chembur	watch		Power + refineries	Mumbai	750	42.6	63.8	0.67	-0.09	yes
Paras	low		Coal power	Akola	500	32.4	13.4		-0.348	planned
Taloja MIDC	low		Industrial cluster	Raigad		31.7	47.7	0.66		
JSW Ratnagiri (Jaigad)	low		Coal power	Ratnagiri	1200	22.8	9.4			
Nashik (Eklahare)	low		Coal power	Nashik	630	15.2	17.4	0.87	-0.294	
Tarapur MIDC	low		Industrial cluster	Palghar		7.9	8.3			
Dahanu (Adani)	low		Coal power	Palghar	500	2.8	10.2			yes

Data sources

Dataset	Provider	Used for	Licence / citation
Sentinel-5P TROPOMI offline L3 SO2 column	ESA / Copernicus	SO2 above local background, per site and month	Copernicus open licence.
Sentinel-5P TROPOMI offline L3 tropospheric NO2 column	ESA / Copernicus	NO2 above local background: how hard the plant runs; denominator of the scrubber ratio	Copernicus open licence.
Global Power Plant Database v1.3	World Resources Institute	Plant locations and capacity (2018 snapshot)	CC BY 4.0.
Chen et al. (2026), super-resolution SO2 emissions over India from TROPOMI	Atmospheric Measurement Techniques 19, 2737	Evidence that plant-level SO2 is visible from TROPOMI (95% of plants above 1,000 MW detected)	CC BY 4.0.
MoEFCC notification on SO2 norms for thermal plants, 11 Jul 2025	Ministry of Environment, Forest and Climate Change	Which plants are exempt from FGD (Category C) or reviewed case by case (Category B)	Public.
Mahagenco annual reports 2023-24 and 2024-25; CSTPS EC compliance report Jan–Jun 2026	Maharashtra State Power Generation Co.	Planned FGD dates for Koradi, Paras, Chandrapur, Bhusawal, Parli	Public.
Plant scrubber status: The Live Nagpur (23 Oct 2026) and Mitsubishi Power on NTPC Mauda; Power Engineering on Dahanu seawater FGD; MPCB consent (2012) for Tata Power Trombay	News and company sources	Reference plants with scrubbers (Mauda, Dahanu, Trombay)	Public.
CPCB Comprehensive Environmental Pollution Index action plans (2020)	Central Pollution Control Board	Critically polluted industrial clusters (Tarapur, Navi Mumbai, Chandrapur)	Public.

Satellite columns show where and when a plant's plume changed. They do not measure stack concentration; use them to choose which plants to inspect and to check whether scrubbers changed what reaches the air.